

Library Management System - Quick Start Guide

Current Status

What's Ready

-  All code files prepared and committed locally
-  Git configured (Arnav Reddy / basketballarms@gmail.com)
-  Dependencies installed
-  Server configured for port 5000
-  Environment variables set up

What Needs Your Action

CRITICAL: MongoDB Atlas IP Whitelisting

Your Server IP: 198.212.42.22

Steps:

1. Go to [MongoDB Atlas](https://cloud.mongodb.com/) (https://cloud.mongodb.com/)
2. Click **Network Access** → **IP Whitelist**
3. Click **Add IP Address**
4. Either:
 - Quick: Click "Allow Access from Anywhere" (adds 0.0.0.0/0)
 - Secure: Enter 198.212.42.22
5. Click **Confirm**
6. Wait 1-2 minutes

OPTIONAL: GitHub Push

The code is ready but couldn't be pushed due to permissions.

Fix:

1. Visit [GitHub App Settings](https://github.com/apps/abacusai/installations/select_target) (https://github.com/apps/abacusai/installations/select_target)
2. Select repository: Collegelibrary_ABACUS1
3. Grant access to Abacus.AI GitHub App

Or manually push:

```
# On your local machine
git clone https://github.com/ArnavDada14/Collegelibrary_ABACUS1.git
cd Collegelibrary_ABACUS1
# Copy files from /home/ubuntu/library-system
git add .
git commit -m "Add complete library management system"
git push origin main
```

Deploy Your Application

After MongoDB IP Whitelisting

Option 1: Automated (Recommended)

```
cd /home/ubuntu/library-system
./deploy_after_whitelist.sh
```

Option 2: Manual

```
cd /home/ubuntu/library-system

# Seed database
npm run seed

# Start server in background
nohup npm start > /tmp/server.log 2>&1 &

# Check logs
tail -f /tmp/server.log
```

Access Your Application

Public URL: <https://11a8ddaba7-5000.na104.preview.abacusai.app>

 **Note:** This URL works while the VM is running. The VM auto-shuts down after inactivity.

Login Credentials

Admin Account

- **Email:** admin@library.com
- **Password:** admin123
- **Access:** Full admin dashboard, manage users, manage books

Student Accounts

- **Email:** student1@library.com
- **Password:** student123
- **Access:** Browse catalog, issue/return books

Additional students: student2@library.com, student3@library.com (same password)

What Gets Created

When you run the seed script:

- **4 Users:** 1 admin + 3 students
- **10 Books:** Sample library catalog including:

- To Kill a Mockingbird
 - 1984
 - The Great Gatsby
 - Pride and Prejudice
 - The Catcher in the Rye
 - And 5 more classics
-



Useful Commands

```
# View server logs
tail -f /tmp/server.log

# Check if server is running
curl http://localhost:5000

# Stop server
pkill -f "node server/server.js"

# Restart server
pkill -f "node server/server.js"
cd /home/ubuntu/library-system
nohup npm start > /tmp/server.log 2>&1 &

# Test MongoDB connection
cd /home/ubuntu/library-system
node -e "require('dotenv').config(); const m=require('mongoose');
m.connect(process.env.MONGODB_URI).then(()=>console.log('✅ Connec-
ted')).catch(e=>console.log('❌ Failed:', e.message))"
```



Features Overview

For Students

- Browse complete book catalog
- Search by title, author, or ISBN
- Filter by genre and availability
- Issue books (max 3 per student)
- Return books
- View your issued books

For Admins

- Everything students can do, plus:
- Add new books to catalog
- Update book information
- Delete books
- View all users
- View all issued books
- Manage book availability

Troubleshooting

Server won't start

```
# Check if port is in use
lsof -i :5000

# Kill existing process
pkill -f "node server/server.js"

# Check logs
tail -f /tmp/server.log
```

MongoDB connection fails

1. Verify IP is whitelisted in Atlas
2. Check `.env` has correct `MONGODB_URI`
3. Wait 1-2 minutes after whitelisting
4. Test connection with command above

Can't access public URL

1. Ensure server is running: `curl http://localhost:5000`
2. Check server logs for errors
3. Verify VM is active

Project Info

- **Repository:** https://github.com/ArnavDada14/Collegelibrary_ABACUS1
- **Local Path:** `/home/ubuntu/library-system`
- **Latest Commit:** `a3c206764653ef7015d461efa7d8315de9960fc3`
- **Branch:** `main`

Next Steps

1. **Whitelist IP** in MongoDB Atlas → <https://cloud.mongodb.com/>
2. **Run deployment script:** `./deploy_after_whitelist.sh`
3. **Access app:** <https://11a8ddaba7-5000.na104.preview.abacusai.app>
4. **Login as admin:** `admin@library.com` / `admin123`
5. **Test features:** Add books, manage users, issue/return books

Need help? Check the detailed documentation:

- `/home/ubuntu/library-system/README.md`
- `/home/ubuntu/library-system/DEPLOYMENT_SUMMARY.md`

- /tmp/deployment_status.md
- /tmp/final_deployment_summary.txt